

SOC ANALYST PORTFOLIO CASE STUDY

Targeted Network Isolation with Wazuh and UFW

Detecting reconnaissance, containing one source, preserving trusted service availability, and validating recovery

A level-7 firewall alert was converted into a reversible source-isolation action while authorized HTTP access remained available.

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Exercise date	August 26, 2026
Environment	Authorized VirtualBox SOC home lab
Classification	Portfolio demonstration - no production systems

AUTHORIZED-LAB SAFETY NOTE

All scanning, authentication attempts, firewall actions, and log collection occurred inside a controlled private network owned and operated by the analyst.

Executive Summary

A controlled Nmap scan from Parrot Security (10.10.10.5) generated logged UFW blocks on LINUX-WEBSERVER01 (10.10.10.4). Wazuh correlated the firewall telemetry through custom rule 100100 at level 7. After confirming the source, destination, and blocked ports, the analyst inserted a source-specific deny rule ahead of normal service allowances. Follow-up testing showed TCP 22, 80, and 8080 as filtered from the simulated source, while an authorized request from the Wazuh server still received HTTP 200. The temporary containment rule was then removed, normal access was restored, and ten evidence files passed SHA-256 verification.

DISPOSITION

Controlled reconnaissance detected and contained. No host compromise, service outage, or data disclosure was confirmed.

Objective and Scope

- **Primary objective:** Demonstrate a SOC workflow that moves from detection to targeted containment and recovery.
- **Detection scope:** UFW block events generated by controlled TCP probes from 10.10.10.5.
- **Containment scope:** Deny new inbound traffic from the identified source without blocking trusted hosts.
- **Validation scope:** Verify the source was filtered, confirm authorized web access remained available, and preserve evidence integrity.

Lab Environment

Role	System	Address / Function
Source	Parrot Security	10.10.10.5 - controlled scanner
Protected endpoint	LINUX-WEBSERVER01	10.10.10.4 - SSH and Apache
SIEM	Wazuh server	10.10.10.3 - alert collection and analysis
Network	VirtualBox NAT Network	10.10.10.0/24 - isolated lab segment

Safety boundary: All systems used private RFC 1918 addresses, and the activity remained inside the analyst-owned home lab.

Detection and Investigation Workflow

Baseline and controlled activity

The protected endpoint was reachable, and baseline reconnaissance showed SSH on TCP 22 and Apache on TCP 80. A controlled top-100 TCP SYN scan was then used to create firewall telemetry without exploiting a service.

```
[*]-[user@parrot]-[~]
-> $sudo nmap -sS -T3 --top-ports 100 --reason 10.10.10.4 -oN "$HOME/SOC-Labs/Network-Isolation/recon/03-controlled-scan.txt"
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-26 17:17 UTC
Nmap scan report for 10.10.10.4
Host is up, received arp-response (0.0032s latency).
Not shown: 98 filtered tcp ports (no-response)
PORT      STATE SERVICE REASON
22/tcp    open  ssh      syn-ack ttl 64
80/tcp    open  http     syn-ack ttl 64
MAC Address: 08:00:27:91:C9:2A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.11 seconds
[user@parrot]-[~]
```

Figure 1. Controlled Nmap scan confirmed the server was online and exposed TCP 22 and 80 before containment.

SIEM triage

Wazuh recorded the blocked probe as rule 100100, level 7, with source 10.10.10.5 and destination 10.10.10.4. The visible event targeted TCP 3389 and showed rule.firedtimes of 20, supporting a reconnaissance pattern rather than a single accidental connection.

agent.name	LINUX-WEBSEVER01
decoder.name	kernel
full_log	Aug 26 17:17:06 linux-webserver01 kernel: [UFW BLOCK] IN=enp0s3 OUT= MAC=08:00:27:91:c9:2a:08:00:27:3b:a4:b2:08:00 SRC=10.10.10.5 DST=10.10.10.4 LEN=44 TOS=0x00 PREC=0x00 TTL=40 ID=37368 PROTO=TCP SPT=58728 DPT=3389 WINDOW=1024 RES=0x00 SYN URG=0
id	1787764627.20605
input.type	log
location	journalid
manager.name	wazuh-server
predecoder.hostname	linux-webserver01
predecoder.program_name	kernel
predecoder.timestamp	Aug 26 17:17:06
rule.description	UFW blocked inbound network connection.
rule.firedtimes	20
rule.groups	local, syslog, sshd, firewall, network_connection_denied
rule.id	100100
rule.level	7
rule.mail	false
timestamp	Aug 26, 2026 @ 13:17:07.353

> Aug 26, 2026 @ 13:17 ⊕ ⊖ predecoder.hostname: linux-webserver01 predecoder.program_name: kernel

Figure 2. Wazuh alert showing rule 100100, level 7, source 10.10.10.5, destination 10.10.10.4, and a blocked TCP probe.

Containment Action

The analyst inserted a logged deny rule at position 1 so it would take precedence over existing subnet allowances:

CONTAINMENT COMMAND

sudo ufw insert 1 deny in log from 10.10.10.5 to any comment 'SOC Lab Incident Isolation'

The rule was source-specific and reversible. It did not alter the server's real SSH or Apache service configuration and did not remove the existing allow rules for the lab subnet.

```
socstudent@linux-webserver01:~$ sudo ufw insert 1 deny in log from 10.10.10.5 to any comment 'SOC Lab Incident Isolation'
[sudo] password for socstudent:
Rule inserted
socstudent@linux-webserver01:~$ sudo ufw status numbered
Status: active

      To      Action      From
      --      -
[ 1] Anywhere  DENY IN    10.10.10.5      (log) # SOC Lab Incident Isolation
[ 2] 8080/tcp  DENY IN    Anywhere        # SOC Lab Block Test
[ 3] 22/tcp    ALLOW IN   10.10.10.0/24   # SOC Lab SSH Subnet
[ 4] 80/tcp    ALLOW IN   10.10.10.0/24   # SOC Lab Apache Subnet
[ 5] 8080/tcp  DENY IN    Anywhere (v6)   # SOC Lab Block Test

socstudent@linux-webserver01:~$
```

Figure 3. The source-isolation rule appears at position 1 ahead of the normal SSH and Apache allowances.

Containment Validation

Three independent checks were used to validate the response:

- **Blocked-source test:** From Parrot, TCP 22, 80, and 8080 all appeared filtered with no response.
- **Trusted-access test:** From the Wazuh server, Apache continued to return HTTP status 200.
- **Telemetry test:** Wazuh continued to receive UFW blocks after isolation, including TCP 22; rule.firedtimes increased to 32 and the filtered search returned 12 recent hits.

```
[user@parrot]~$ sudo nmap -sS -Pn -p 22,80,8080 --reason 10.10.10.4 -oN "$HOME/SOC-Labs/Network-Isolation/evidence/05-post-isolation-scan.txt"
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-26 17:53 UTC
Nmap scan report for 10.10.10.4
Host is up, received arp-response (0.0032s latency).

PORT      STATE      SERVICE      REASON
22/tcp    filtered  ssh          no-response
80/tcp    filtered  http         no-response
8080/tcp   filtered  http-proxy   no-response
MAC Address: 08:00:27:91:C9:2A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.34 seconds
[user@parrot]~$
```

Figure 4. Post-isolation Nmap validation showed TCP 22, 80, and 8080 as filtered from 10.10.10.5.

t agent.name	LINUX-WEBSEVER01
t decoder.name	kernel
t full_log	Aug 26 17:53:51 inux-webserver01 kernel: [UFW BLOCK] IN=enp0s3 OUT= MAC=08:00:27:91:c9:2a:08:00:27:3b:a4:b2:08:00 SRC=10.10.10.5 DST=10.10.10.4 LEN=44 TOS=0x00 PREC=0x00 TTL=42 ID=19836 PROTO=TCP SPT=40774 DPT=22 WINDOW=1024 RES=0x00 SYN URG=0
t id	1787766833.63063
t input.type	log
t location	journalld
t manager.name	wazuh-server
t predecoder.hostname	inux-webserver01
t predecoder.program_name	kernel
t predecoder.timestamp	Aug 26 17:53:51
t rule.description	UFW blocked inbound network connection.
# rule.firedtimes	32
t rule.groups	local, syslog, sshd, firewall, network_connection_denied
t rule.id	100100
# rule.level	7
rule.mail	false
timestamp	Aug 26, 2026 @ 13:53:53.992

Figure 5. Wazuh recorded continued blocked traffic after containment; the displayed event targeted TCP 22 and rule 100100 had fired 32 times.

Recovery and Evidence Integrity

After validation, the analyst removed only the temporary rule. Follow-up scanning confirmed TCP 22 and 80 were open again, demonstrating a controlled rollback. The evidence collection contained ten files, including baseline tests, scan results, UFW logs, rule status, timestamps, and post-isolation validation. Every recorded SHA-256 hash verified successfully.

```
[user@parrot]~[~/SOC-Labs/Network-Isolation]
$ sudo nmap -sS -Pn -p 22,80 --reason 10.10.10.4
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-26 19:15 UTC
Nmap scan report for 10.10.10.4
Host is up, received arp-response (0.0041s latency).

PORT      STATE SERVICE REASON
22/tcp    open  ssh     syn-ack ttl 64
80/tcp    open  http    syn-ack ttl 64
MAC Address: 08:00:27:91:C9:2A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.20 seconds
[user@parrot]~[~/SOC-Labs/Network-Isolation]
$
```

Figure 6. After the temporary rule was removed, TCP 22 and 80 were reachable again.

```

[user@parrot]--[~/SOC-Labs/Network-Isolation]
$ sha256sum -c evidence/SHA256SUMS.txt
./evidence/05-post-isolation-scan.txt: OK
./evidence/network-isolation-post-block-evidence.log: OK
./evidence/network-isolation-rule-status.txt: OK
./evidence/network-isolation-ufw-evidence.log: OK
./notes/lab-end.txt: OK
./notes/lab-start.txt: OK
./recon/01-baseline-connectivity.txt: OK
./recon/02-connectivity-retest.txt: OK
./recon/03-controlled-scan.txt: OK
./recon/04-denied-port-test.txt: OK
[user@parrot]--[~/SOC-Labs/Network-Isolation]
$

```

Figure 7. All ten Network Isolation evidence files passed SHA-256 verification.

Evidence-to-Conclusion Matrix

Evidence	Observed fact	Analyst conclusion
Baseline scan	TCP 22 and 80 open	Services were reachable before containment
Wazuh alert	Rule 100100, level 7, SRC 10.10.10.5	Firewall blocked controlled reconnaissance from the identified source
UFW status	Source deny rule at position 1	Containment took precedence over normal allow rules
Post-isolation scan	22/80/8080 filtered	The simulated source could no longer reach tested services
Authorized curl	HTTP 200 from 10.10.10.3	Containment remained selective rather than causing a general outage
Hash verification	10 files reported OK	Collected artifacts were unchanged after hashing

Findings and Limitations

Confirmed findings

- The controlled source generated repeated TCP probes that were blocked by the endpoint firewall.
- Wazuh successfully converted UFW telemetry into a level-7 detection using custom rule 100100.

- A source-specific deny rule blocked the tested services from 10.10.10.5 while trusted HTTP access remained available.
- The temporary response was reversible, and normal service reachability returned after rollback.

What was not proven

- No exploitation, authentication bypass, malware execution, persistence, or data exfiltration occurred.
- A filtered Nmap result demonstrates the scanner's perspective; it does not independently prove the internal state of every service.
- The exercise used one small virtual network and did not test enterprise routing, redundant firewalls, or large-scale alert volume.

Recommendations

Priority	Recommendation	Rationale
High	Use time-bound, source-specific active-response rules with approval and rollback controls.	Reduces exposure while limiting unintended service disruption.
High	Correlate UFW blocks with authentication, endpoint, and asset-criticality telemetry.	Separates noisy scanning from activity that indicates real compromise risk.
Medium	Validate containment from at least one blocked and one trusted network location.	Proves both enforcement and continued business availability.
Medium	Retain rule-change records and cryptographic evidence manifests.	Improves auditability and incident reconstruction.
Low	Tune repetition thresholds and suppression windows for recurring scans.	Reduces alert fatigue without losing source-level visibility.

Troubleshooting and Engineering Judgment

- Distinguished a Wazuh sudo-command audit event from the intended firewall detection by filtering specifically on rule.id 100100.
- Placed the isolation rule ahead of service allowances so rule order could not silently defeat containment.
- Preserved console access and validated trusted HTTP availability before calling the action successful.
- Recovered a temporarily frozen Parrot VM without saving the failed memory state; all previously written evidence survived.
- Used SHA-256 validation before removing the response rule and restoring the environment.

Employer-Facing Project Summary

This project demonstrates an end-to-end analyst response: validate an alert, identify the source and affected asset, choose a minimally disruptive containment action, verify both security enforcement and service availability, preserve evidence, and perform a controlled rollback.

Resume Bullet

PORTFOLIO-READY BULLET

Investigated Wazuh/UFW reconnaissance alerts, isolated a confirmed source with a precedence-aware firewall rule, validated targeted blocking while preserving authorized HTTP service, and verified 10 evidence artifacts with SHA-256.

Interview Talking Points

- Why a source-specific rule was chosen instead of disabling the interface or shutting down the server.
- How the trusted HTTP 200 check prevented a false claim of successful containment that actually caused an outage.
- Why Wazuh repetition counts and raw UFW fields were treated as evidence, while compromise was explicitly not claimed.

Skills Demonstrated

SOC alert triage | Wazuh custom rules | Linux UFW | Nmap validation | Containment and recovery | Evidence integrity | Analyst documentation

AI and Source Transparency

The analyst personally configured the virtual machines, executed the commands, validated the alerts, interpreted the results, captured the screenshots, and preserved the evidence. Generative AI assisted with command sequencing, troubleshooting suggestions, and documentation structure. Technical claims were checked against direct lab evidence and the official references listed below.

References

- Ubuntu Server documentation - Firewall. <https://ubuntu.com/server/docs/how-to/security/firewalls/>
- Wazuh documentation - Custom rules. <https://documentation.wazuh.com/current/user-manual/ruleset/rules/custom.html>
- Nmap Reference Guide - Port Scanning Basics. <https://nmap.org/book/man-port-scanning-basics.html>